

Defusing Affective Polarization with a Co-Authored Community Note: A Field Experiment with Rival K-pop Fandoms

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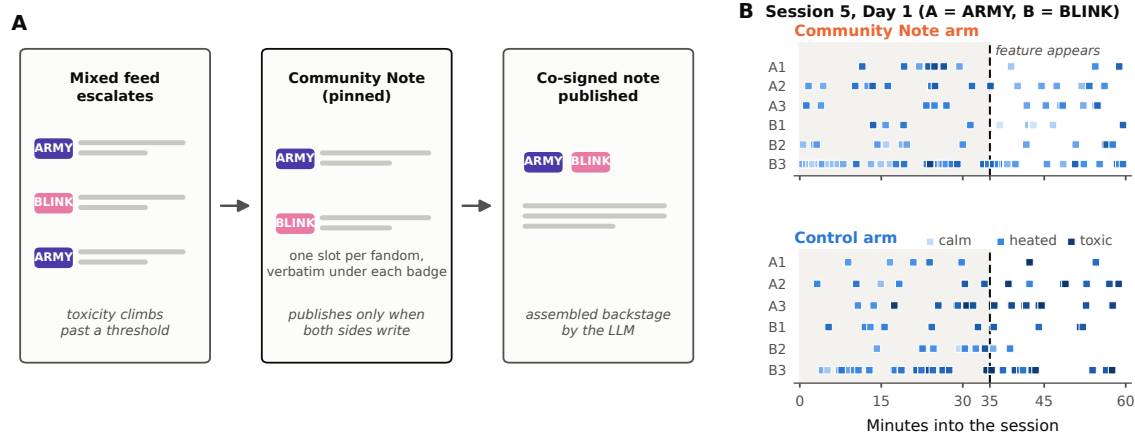


Fig. 1. A ceasefire, not a peace treaty. (A) When toxicity in a mixed ARMY-BLINK feed crosses a threshold, the platform pins a Community Note with one contribution slot per fandom; the note publishes only when both sides have written their part, each entry verbatim under its author's badge, assembled backstage by an LLM that never appears as a participant. (B) One Day-1 session in both arms, one square per message, shaded from calm to toxic: six fans per arm (rows A1-A3 ARMY, B1-B3 BLINK) over 60 minutes, with the assigned feature appearing at 35 minutes (dashed line). In the Community Note arm (107 messages) the squares stay light after the line; in the control arm (91 messages) they darken. The session shown is the one with the median treatment effect of the five, not the strongest.

Hostility between online communities is rooted in group identity rather than substantive disagreement, and interventions aimed at beliefs leave it untouched. Drawing on cooperative interdependence theory, we built a platform feature through which rival groups must co-author a single co-signed post: a Community Note with one slot per group that publishes only when both sides have written. We deployed it on KFeed, our forum, in five two-day sessions where members of ARMY and BLINK, rival fandoms of BTS and BLACKPINK, discussed K-pop; matched control sessions received an inert feature at the same trigger. Control sessions escalated after the trigger; note sessions did not, a 25.3-point difference that held in every session and persisted the next day without the feature. The calm came from the periphery: the heaviest posters grew hotter. We argue the note interrupts the attention escalation requires, and discuss interventions that need not be accepted to work.

CCS Concepts: • **Human-centered computing** → **Collaborative and social computing**.

Additional Key Words and Phrases: Affective Polarization, Community Notes, Online Communities, Intergroup Contact, Large Language Models, Fandom

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1 INTRODUCTION

People like to fight online over who they are, not what to do. This hostility toward out-groups, rooted in group identity rather than substantive disagreement, is known as affective polarization [16, 17], and it pervades online communities [29, 36]. Common intuitive solutions target what people think. However, neither works reliably: exposure to the other side can entrench positions [2], and belief-correcting interventions produce modest, short-lived effects [38]. If the animosity is anchored in identity, interventions aimed at information miss the layer where the conflict lives.

Classic intergroup research suggests that hostility is reduced not by changing beliefs, but by changing relationships between groups. When rival groups become mutually dependent on a shared goal, cooperation fosters a common identity and reduces conflict [12, 30]. This mechanism, cooperative interdependence, has shown robust effects in real-world settings [21, 26]. However, existing online depolarization interventions target individuals rather than group relationships [8, 14, 41], and none brings rival groups into cooperative production. In this study, we investigate whether structured co-creation between rival groups can reduce intergroup hostility on online platforms.

Online fights are rarely intense from the first message. They accumulate: a topic ignites them, and they burn only as long as the participants keep feeding them attention. That dependence is the opening we exploit. We instantiate the idea as Community Notes, a platform feature through which members of rival groups co-author a single, publicly co-signed post—a task that asks nothing of anyone’s beliefs but takes a few minutes of everyone’s attention away from the fight. A pinned prompt offers one slot per group; the note publishes only when both sides have written their part, each entry verbatim under its author’s badge. Neither side can complete a note alone, and neither has to stop being itself. The LLM behind the feature stays out of sight: it monitors the discussion and publishes the finished note. Participants encounter a platform feature and each other, not a machine.

Testing such a feature requires rival groups whose hostility is real, identity-driven, and native to platforms. We chose K-pop fandom, one of the largest transnational fan cultures online [19, 24], and the rivalry between ARMY and BLINK, fandoms of BTS and BLACKPINK: it is enduring, conducted almost entirely on platforms, and rests on no substantive dispute, a near-pure case of identity-driven hostility [10, 42]. We deploy the feature on KFeed, a Reddit-style forum built for this research, where mixed sessions of ARMY and BLINK members discuss K-pop topics with either the Community Note or an inert control feature. We measure toxicity through automated scoring and human coding, and qualitatively analyze how cross-group interaction changes. We ask:

- **RQ1:** How do Community Notes affect polarized groups posting on online forums?
- **RQ2:** How does what they post change after the Community Note intervention?

In answering these questions, our work contributes: (1) a reusable platform mechanism that operationalizes cooperative interdependence in an online forum; (2) an extension of LLM-based depolarization from persuading individuals to structuring how groups interact; and (3) empirical evidence from real polarized communities that a shared task can call a ceasefire without producing a peace: sessions with the note did not escalate while matched control sessions did, the difference persisted a day later with no feature on screen, and the calm came from the periphery of the fight rather than its core.

2 RELATED WORK

Defusing polarization in online forums raises three questions: what kind of polarization it is, what reduces it, and what role an AI system can play. We first characterize affective polarization as a group-level phenomenon rooted in social identity. We then draw on cooperative interdependence theory to elicit the requirements for our intervention. Finally, we examine how LLMs have been used to intervene in online communities.

2.1 Affective Polarization in Online Fan Communities

Affective polarization refers to animosity toward an out-group that is rooted in group membership rather than substantive disagreement [16, 17]. Social identity theory traces this animosity to the self-concept: because people define themselves partly through their groups, favoring the in-group and derogating the out-group becomes a way of maintaining a positive self-image [34]. Strikingly, this requires almost nothing. In minimal group experiments, assigning people to arbitrary, meaningless categories was enough to produce discrimination between groups [33]. Hostility follows from the group boundary itself, not from any dispute across it. The intuitive remedy, correcting what each side believes about the other, therefore addresses the wrong layer. Polarization is a membership problem, not an information problem.

Political contexts make this hard to see, because partisan identity is entangled with real ideological stakes and the contribution of identity alone cannot be isolated [23]. Online fan communities present the phenomenon in purer form. Rival fandoms sustain hostility toward one another with no substantive dispute to resolve [10]. The bond runs deeper than preference: fans incorporate their idol’s successes and failures into their own self-worth, basking in reflected glory when the idol prevails [5], and the more strongly they identify, the less able they are to distance themselves when the idol is attacked [39].

This self-investment explains why fan conflict takes the form of costly, active aggression rather than mere dislike. Among the most devoted fans, personal and group identity can fuse, and fused individuals do not simply favor their group: they act for it, willingly bearing personal costs to defend it [31, 32]. Notably, this conflict unfolds within platform mechanics: fans fight with posts, downvotes, and coordinated campaigns, in forums and comment sections, at times over the use of those spaces themselves [22, 42]. Perceived threats to the group’s standing harden identification further rather than loosening it [3]. In idol and sports forums, toxicity and hostile language accordingly concentrate around encounters with rival groups [6, 29]. K-pop fandoms are a prominent case, organized, transnational, and highly active online [19, 24].

HCI has largely overlooked this side of fandom: the field’s fan research centers on positive collective action [11], and fandom itself remains a self-acknowledged gap [11]. Fan communities thus offer what political settings cannot: a near-pure case of identity-driven animosity, unfolding entirely within platform mechanics, and a clean setting for testing interventions that target identity rather than information.

2.2 Cooperative Interdependence and Intergroup Contact

The intuitive remedies for intergroup hostility are more contact and better information. Neither reliably changes identity-based conflict. Exposing partisans to opposing views entrenched their positions rather than softening them [2], and a megastudy of depolarization interventions found even the strongest effects modest and short-lived [38]. Yet one apparent failure reveals a design opportunity. In a field experiment, Iraqi Christians were assigned to amateur soccer teams that were either all-Christian or mixed with Muslim teammates [26]. Playing together did not change their general attitudes toward Muslims, but it changed their behavior. Months later, players from mixed teams were

more likely to train with Muslim peers and sign up for mixed leagues. Attitudes rooted in identity are difficult to shift directly, while cooperative behavior emerges when groups share a goal that neither side can achieve alone.

This mechanism is known as cooperative interdependence. In the Robbers Cave experiments, contact alone intensified competition between rival groups, until they were placed in situations requiring collaboration toward superordinate goals [30]. The Common Ingroup Identity Model explains why. Working toward a shared outcome leads people to recategorize themselves under a broader identity, extending the positive regard reserved for the in-group to former out-group members [12]. The mechanism operates among real-world rivals. Manchester United fans who encountered an injured jogger were more likely to help when he wore a United shirt than a Liverpool one; when a shared identity as football fans had been made salient beforehand, they helped the Liverpool supporter just as readily [21]. In both cases, behavior changed while group identities stayed intact.

These findings set three requirements for interventions targeting identity-driven conflict. Interaction must involve genuine interdependence, since tasks that either group can complete alone provide no motivation for recategorization [12]. The shared identity must be layered on top of existing subgroup identities rather than replacing them, because attempts to erase group boundaries read as identity threats, which intensify rather than reduce identification [3, 15]. And the shared identity must emerge from the interaction itself rather than being imposed explicitly, because categorization follows the identities made salient by the immediate situation [21, 37]. Our intervention is built to satisfy all three.

Online platforms are deeply involved in shaping these interactions. Platform affordances enable rival communities to organize, amplify, and sustain hostility at scale [22], and platform design can further intensify rivalry among fandoms [42]. But the same mechanisms that structure conflict can be redesigned to support cooperation. Existing approaches intervene at three points, setting aside moderation, which suppresses hostile expression without changing the relationship between groups. Exposure-based designs modify what users encounter across group boundaries, such as cross-cutting feeds or shared casual games [2, 14]. Matching-based designs modify who interacts with whom, pairing individuals across the divide [8]. Correction-based designs modify how contested information is evaluated, most notably Community Notes, where a bridging mechanism surfaces contributions considered helpful by users with otherwise opposing viewpoints [41].

These approaches leave the relationship between the groups unchanged. Polarized communities remain audiences exposed to one another, strangers temporarily paired, or adversaries evaluating each other's claims. None intervenes at the point where relationships are formed through joint creation, where opposing groups must produce something together rather than merely encounter, discuss, or judge one another. Cooperative interdependence, the mechanism with the strongest field evidence for changing intergroup behavior, has never been built into a platform and tested with members of real polarized communities.

2.3 LLM Interventions in Online Communities

When AI systems intervene in online conflict, they take one of the roles that conflict resolution research assigns to human third parties. Moderators enforce conduct rules after violations occur, facilitators manage participation without touching content, and mediators contribute questions and proposals of their own [25]. LLMs have now filled each seat. They moderate at scale [13], suggest real-time rephrasings that make contentious conversations feel heard [1], reduce belief in conspiracy theories through persuasive dialogue [9], and mediate live debates with psychologically grounded strategies [35]. Across this spectrum, what varies is how much the system contributes to the conversation.

Two features hold constant. First, the intervention targets individual minds. Each system persuades, corrects, or calms particular interlocutors, treating polarization as something wrong in what a person thinks. The results echo a

familiar pattern. LLM dialogue can move issue positions, yet affective polarization stays where it was [9, 35], the same dissociation Mousa observed on the soccer field, and the one expected if hostility follows membership rather than belief. Second, the system is a visible party to the exchange. Users know they are talking with, or being coached by, a machine, and disclosed AI authorship invites discounting and reactance [18]. The deeper problem is theoretical. An agent that openly steers people toward common ground is doing precisely what the third requirement rules out, announcing a shared identity rather than letting the situation produce one.

Across this spectrum, then, the LLM holds a seat in the interaction, and users are acted upon by the machine. No existing work inverts that allocation, returning the interaction entirely to the users while confining the system to the conditions under which they meet. In that position the LLM shapes when cooperation becomes possible and how a joint product takes form, but what participants encounter, throughout, is each other.

Together, the three literatures leave a single gap. Affective polarization has been studied almost entirely in political settings, leaving its purest identity-driven form in fan communities unexamined [10, 11]. Cooperative interdependence, the mechanism with the strongest field evidence, has never been built into a platform [26, 30]. And LLM interventions have varied the system’s role within the conversation without asking whether it belongs in the conversation at all [1, 35]. Our work sits at this intersection. We build a community notes feature through which members of rival fandoms co-author a single, publicly co-signed post, with the LLM working only backstage, and test it with two polarized communities on a live forum.

3 EXPERIMENT: COMMUNITY NOTES IN A POLARIZED FORUM

We conducted a two-day experiment to test whether a co-authorship feature reduces toxicity between rival fandoms. Participants, self-identified members of ARMY or BLINK, joined mixed sessions of ARMY and BLINK members on KFeed, a Reddit-style forum built for this research, and discussed K-pop topics under their fandom badge. Discussion threads were seeded with prompts, including toxic arguments drawn from pilot sessions, so that hostile, neutral, or constructive interaction could emerge naturally.

Throughout each session, the platform scored incoming posts for toxicity in real time. When session-level toxicity crossed a threshold determined from pilot sessions, the assigned feature appeared in the feed; in the sessions reported here this happened 35 minutes into every Day-1 session in both arms, by which point toxicity had climbed past the threshold in each. **[TODO: confirm whether the trigger was threshold-driven or fixed at the 35-minute mark in the deployed protocol, and say which.]** We randomly assigned sessions to two conditions:

- (1) **Community Notes treatment:** the feature that appeared was a Community Note, a pinned prompt framed at the K-pop level with one contribution slot per fandom. The note published only when both sides had written their part, each entry preserved verbatim under its author’s badge.
- (2) **Inert feature control:** the feature that appeared was matched to Community Notes on visual salience and interaction cost but required no cross-fandom contribution. **[TODO: describe the inert feature as fielded.]** Following the placebo-activity logic of Combs et al. [8], this condition ensures that both arms experience a triggered platform event, so that novelty and regression toward the mean are shared across conditions and the treatment effect is identified by the condition-by-phase contrast.

On Day 2, participants returned to the same session composition and discussed without any feature, allowing us to examine whether behavioral changes persisted after the mechanism was removed.

3.1 Building KFeed and the Community Notes Feature

KFeed supports community feeds, threaded discussion, moderator-pinned posts, and persistent fandom badges on every post. The Community Notes feature translates the three requirements from Section 2.2 into interface rules (Figure 1A).

- **A note needs both fandoms.** After one side submits, the entry sits waiting under its author’s badge until a rival answers. Neither side can write for the other or publish alone.
- **Both badges stay visible.** From draft to publication, each entry carries its author’s fandom badge, and votes accrue to the note as a whole rather than to either half.
- **Nothing is said about shared identity.** Prompts address K-pop fans at large, but no instruction tells participants to identify as one group, and entries are published verbatim rather than merged.

3.2 Configuring the Backstage LLM

An LLM performs two functions in the system, both invisible to participants. First, it monitors the discussion: incoming posts are scored for toxicity in real time, producing the session-level signal that triggers the assigned feature. Second, it publishes the Community Note: once both fandom slots are filled, it verifies that the note contains one entry per fandom and assembles the two entries verbatim into the pinned format. The model does not write or rewrite anyone’s words, does not converse with participants, and claims no identity. We keep it backstage deliberately: an agent that visibly steered rivals toward common ground would announce the shared identity that the design requires the situation to produce. In our system, the LLM decides when the cooperative task appears and makes its outcome public; the cooperation itself belongs entirely to the participants.

3.3 Recruitment and Participants

We recruited participants through fandom communities on social platforms, screening with an open-ended fandom identification question to avoid signaling the study’s focus, with working language and location as eligibility criteria. 60 participants took part in the sessions reported here, 6 per session and arm (three ARMY and three BLINK), each attending both days; no participant appeared in more than one session.

3.4 Procedure

Participants completed a pre-survey at recruitment (fandom identification, trait covariates, and eligibility screening), then joined fixed time slots and confirmed readiness before each session. Sessions lasted approximately one hour per day on two consecutive days. A survey followed each day’s session: after Day 1, the outcome battery; after Day 2, the outcome battery again, feature-specific items, and a funneled suspicion probe. After the Day 2 survey, we invited a subset of participants to semi-structured interviews about their experience across both days, sampling from both fandoms and both conditions.

3.5 Measures

Toxicity is the primary outcome, scored on pre-moderation text by Perspective API [20] and independent human coding following Coe, Kenski, and Rains’s incivility codebook [7], supplemented by a K-pop incivility lexicon validated before use with inter-rater reliability reported. The platform stores the result as a 0–100 score per message, which we report on that scale. **[TODO: state how the platform’s 0–100 score is composed: which Perspective attributes, how they are weighted, and how the human codes and lexicon enter the final score.]** Each message in the

Table 1. Day 1 by arm and phase: the first 35 minutes (before the feature) against the remainder (after). The arms are alike before the feature; afterwards the control arm escalates and the note arm does not.

	Community Note arm		Control arm	
	Before	After	Before	After
Messages	272	192	259	177
Mean toxicity (SD)	45.9 (20.1)	44.2 (25.7)	46.9 (20.2)	68.8 (17.9)
Messages scoring ≥ 60	29%	37%	30%	77%
Messages per minute	1.55	1.54	1.48	1.42
Rival fandom as “they” (per message)	0.42	0.19	0.48	0.44
Rival fandom by name (per message)	0.11	0.29	0.12	0.12
Messages about the note task	0	32%	0	10%

export additionally carries three codes applied by the research team: whether it refers to the rival fandom and, if so, whether as “they” or by name; whether it addresses the note task rather than the thread it was posted in; and a flag marking the heaviest poster of each session and arm. **[TODO: report who applied the message codes and their inter-rater reliability.]** Survey outcomes adapt validated instruments to the fandom boundary, following the battery of Rajadesingan et al. [28]: the feeling thermometer toward rival-fandom members [17], common in-group identity representation as the mediator [12], and fandom identification via a fan identification scale [40], which also tests that the intervention does not erode subgroup identity. Behavioral logs capture cross-fandom references and pronoun use [27]. Reactance serves as a backfire guardrail [4].

3.6 Analysis

For RQ1, we compare toxicity across conditions and phases. Because both conditions are triggered at the same point, the treatment effect is identified by the condition-by-phase contrast rather than by a before–after comparison: we report it as a difference-in-differences over sessions (the control arm’s change from before to after the feature minus the note arm’s, one value per session, tested against zero) and, at the message level, as the condition-by-phase interaction in a linear model. The session is the unit of inference throughout; participant- and message-level tests are reported alongside for the reader’s calibration, not as independent evidence. Day 2 tests whether changes persist. For RQ2, we use the message codes (rival-fandom references, pronouns, and note-task topic) and two researchers will code forum posts and interview transcripts for shifts in cross-group interaction.

4 RESULTS

We report 5 two-day sessions, each run in both arms with a fresh mixed group of 6 ARMY and BLINK members, for 60 participants and 1829 messages (900 on Day 1, 929 on Day 2; between 78 and 107 messages per session and arm on Day 1). On Day 1 the assigned feature appeared 35 minutes into each session; on Day 2 the same participants discussed without it. Survey results draw on the 10 treatment-arm participants of the pilot cohorts (AUG16 and TEST1) who completed the post-session battery, none of whom was removed by our two pre-registered screens (straightlining across the battery: 0 excluded; rating the session as not heated, our manipulation check: 0 excluded, minimum rating 4/5). **[TODO: the battery for the five sessions reported here has not been exported; the survey figures below are from the pilot cohorts and must be replaced or explicitly labelled as pilot data.]** Table 1 summarizes Day 1 by arm and phase.

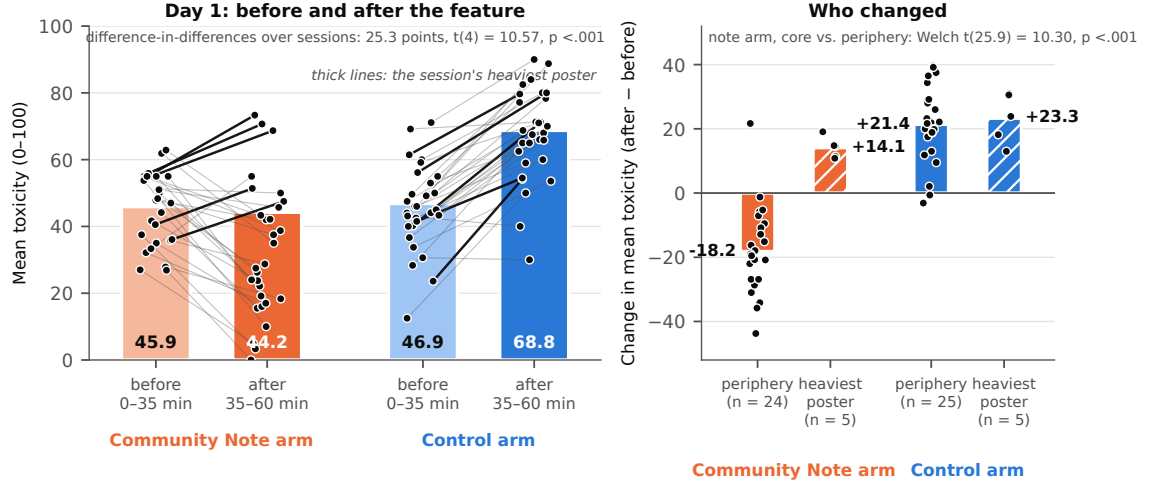


Fig. 2. The escalation the note prevented, with both the averages and the people behind them. Left: Day-1 mean toxicity before and after the feature in each arm (bars: message-level means; dots: participants, connected across the two phases; thick lines: the heaviest poster of each session). Right: each participant's change from before to after, split into the platform-coded heaviest poster of each session and everyone else. In the note arm the periphery cooled while the core heated; in the control arm both rose together.

4.1 The Note Prevented the Escalation the Control Arm Shows (RQ1)

4.1.1 Both arms were alike, and climbing, before the feature. In the 35 minutes before the feature the two arms were indistinguishable: mean message toxicity was 45.9 in the note arm and 46.9 in the control arm (Welch $t(527.5) = -0.53$, $p = 0.593$; session-paired $t(4) = 0.06$, $p = 0.952$). Neither arm was calm: toxicity rose by about a point a minute in both (note arm slope 1.16 per minute, control 0.99, both $p < .001$), from around 29.1 in the first five-minute window to 64.0 and 65.1 in the window just before the feature (Figure 4). The trigger fired, in other words, into a fight that was still heating up.

4.1.2 After the feature, the control arm escalated and the note arm did not. In the control arm, toxicity rose from 46.9 before the inert feature to 68.8 after it (Welch $t(405.8) = 11.90$, $p < .001$), and it rose in every session, by between 15.0 and 28.5 points. In the note arm it went from 45.9 to 44.2 ($t(346.4) = -0.76$, $p = 0.446$), with session changes between -4.1 and 0.4 points. The treatment effect is the difference between these two changes: 25.3 points over sessions (SD 5.4, range 19.1 to 31.2; $t(4) = 10.57$, $p < .001$), and in the message-level model a condition-by-phase interaction of -23.6 points (SE 2.9, $t(896) = -8.26$, $p < .001$). Leaving out any one session moves the difference only between 23.9 and 26.9 points. Figure 2 shows the four means with every participant behind them.

The result does not depend on who is counted (Table 2). Restricting the analysis to participants who posted at least six or at least ten messages leaves the difference-in-differences between 23.1 and 24.8 points; excluding the heaviest poster of each session raises it to 40.4 points, for a reason the next section explains.

4.1.3 The note did not end the discussion. Messages kept arriving in the note arm at 1.54 per minute after the feature against 1.55 before it (rate ratio 0.99, exact test of equal rates $p = 0.925$), the same as in the control arm (1.48 to 1.42). 29 of the 30 note-arm participants posted in both phases, and the one who had been silent before the note joined in after it. The ceasefire cooled the exchange without emptying the room.

Table 2. The condition-by-phase difference survives restricting the sample. Session-level means before and after the feature by arm, and the difference-in-differences (control change minus note-arm change) tested over the five sessions.

Participants included	<i>n</i>	Note arm	Control arm	Difference-in-differences	
		before → after	before → after	points	<i>t</i> (4) (<i>p</i>)
All	60	46.1 → 43.8	45.8 → 68.8	25.3	10.57 (<.001)
≥ 6 messages	56	46.0 → 44.3	45.7 → 68.8	24.8	11.19 (<.001)
≥ 10 messages	44	46.6 → 46.6	45.7 → 68.9	23.1	8.64 (<.001)
Heaviest poster excluded	50	44.8 → 26.9	46.9 → 69.5	40.4	10.52 (<.001)

4.2 The Periphery Cooled and the Core Heated (RQ1)

4.2.1 The flat arm mean hides two opposite movements. At the participant level, the note arm did fall: for the 29 participants who posted in both phases, mean toxicity went from 45.7 to 33.0, a within-participant change of -12.6 points (paired $t(28) = -3.88$, $p < .001$, $d_z = 0.72$), and 23 of the 29 became less toxic. In the control arm the change was 21.7 points ($t(29) = 10.77$, $p < .001$, $d_z = 1.97$), and 28 of 30 became more toxic. The participant-level decline in the note arm is larger than its message-level one because the participants who rose wrote more of the messages.

Those participants were the core of the fight. The platform flags the heaviest poster of each session and arm; these 10 people wrote 42% of all Day-1 messages, and 9 of the 10 were BLINK. In the note arm, all five heavy posters became *more* toxic after the note, from 48.2 to 62.3 on average (changes of 10.8 to 19.1 points), while the other 24 participants went from 45.2 to 26.9 (23 of 24 declined; core versus periphery, Welch $t(25.9) = 10.30$, $p < .001$). The heavy posters' share of the arm's messages also rose, from 42% to 46%. In the control arm core and periphery moved together (14.1 versus -18.2 points in the note arm; 23.3 versus 21.4 in the control arm, $p = 0.663$; Figure 2, right). The note did not lower everyone. It split the room.

4.2.2 The distributions show the split. Figure 3 shows the message-level distributions. Before the feature the two arms overlap. Afterwards the control arm's distribution moves as a block to the right: 77% of its messages score 60 or above, against 30% before. The note arm's distribution instead spreads out, its standard deviation rising from 20.1 to 25.7: most of its mass moves left, and a smaller mass, the heavy posters' messages, moves right, leaving 37% of messages at 60 or above.

4.3 The Conversation Moved to the Task (RQ2)

4.3.1 A third of the note arm's messages were about the note. After the feature, 32% of the note arm's messages (61 of 192) addressed the note task rather than the thread they were posted in, against 10% in the control arm (18 of 177; Fisher's exact $p < .001$). Those task messages were calmer than the rest (37.8 versus 47.3; Welch $t(108.7) = -2.34$, $p = 0.021$), but the more telling number is the other one: the note arm's messages that stayed on the original threads averaged 47.3, only 1.3 points above the arm's pre-feature mean, where the control arm's rose to 68.8. The note did not just siphon off calm messages; the fight it left behind did not heat up either.

4.3.2 They became names. References to the rival fandom as "they" fell by 54% in the note arm, from 0.42 to 0.19 per message (Fisher's exact $p < .001$), while references to the rival fandom by name rose from 0.11 to 0.29. The share of messages referring to the rival fandom at all barely moved (0.53 to 0.48). In the control arm "they" stayed where it was

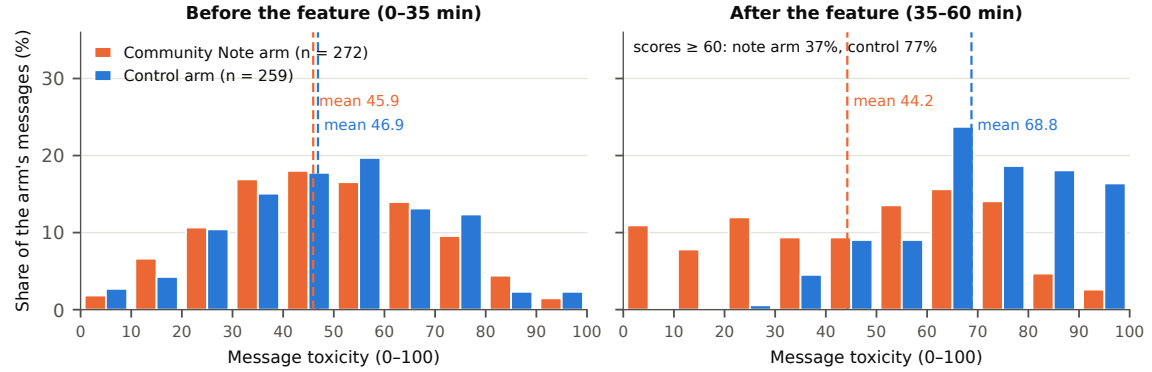


Fig. 3. Distribution of message toxicity before and after the feature, by arm (share of each arm’s messages per 10-point bin). Before the feature the arms overlap. Afterwards the control distribution moves as a block to the right, while the note arm’s spreads: its bulk moves left and a smaller mass, the heavy posters, moves right.

(0.48 to 0.44, $p = 0.494$) and so did naming (0.12 to 0.12). Participants in the note arm kept talking about the other side about as often; what changed is that they named it instead of pointing at it.

4.3.3 Conflict moved from works to persons to identity. Focus-group participants described a similar escalation pattern across discussions. Conflicts often began with disagreements about works, such as comebacks, album sales, view and like counts on YouTube, or choreography. They then shifted to comments about the idol, followed by comments about the other participant, and finally to broader statements about fandom identity: “...when they [BLINKs] talk about BTS, they would say that oh, we ARMY have a fetish of ugly men. They would say we have a problem with aesthetics.” (P#, focus group). [TODO: replace P# with the participant code.] Participants described the note’s interruption as briefly taking them out of the argument, a state one described as *tuozhan*, or standing down: not agreement, but stepping away from that particular fight. [TODO: add supporting quote from transcript.]

Together, these results show that the conversation changed at the same time as the escalation stopped. We do not claim that the topic shift caused it.

4.4 The Ceasefire Minute by Minute

Figure 4 shows both days in 5-minute windows, pooled over the five sessions with each session drawn faintly behind. On Day 1 the two arms climb together for 7 windows. At the feature they part: the control arm continues to 60.7 in the first window after the feature and 75.5 in the last, while the note arm drops to 47.3 and then 31.7 (40–45 minutes), half of the 64.0 it had reached in the window before the note (Welch $t(47.4) = 6.65$, $p < .001$; we report this peak-to-trough contrast as descriptive, since its windows are chosen after the fact; the control arm reads 65.1 and 72.6 in the same two windows). The note arm then drifts back up over the remaining windows (58.2 in the last; post-feature slope 0.60 per minute, $p = 0.014$), so that by the end of Day 1 the gap between the arms had narrowed to 17.3 points. Within the hour, the pause was decaying.

4.5 The Ceasefire Held on Day 2

On Day 2 the same participants returned and discussed without any feature. The note arm’s sessions averaged 44.5 (range 41.2 to 48.1), no different from their Day-1 post-feature level (43.8; paired $t(4) = 0.30$, $p = 0.777$) or from their

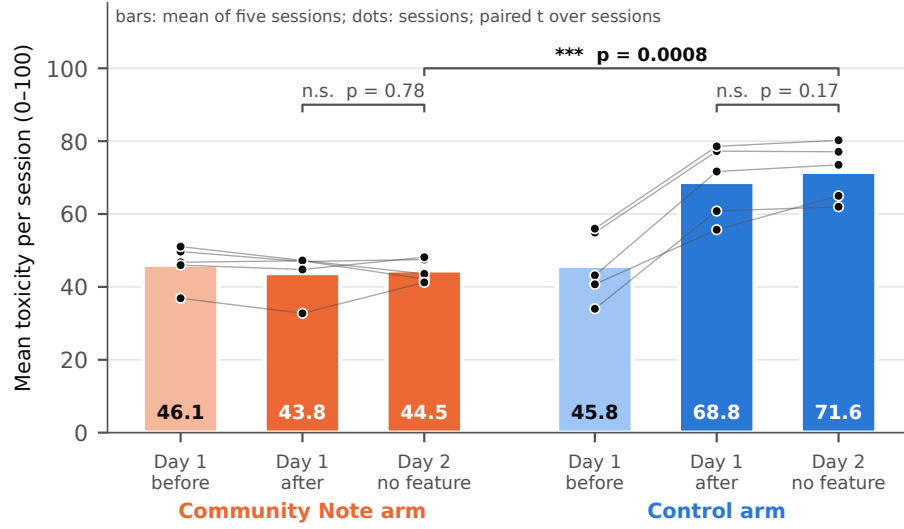


Fig. 5. Session-level mean toxicity in the Day-1 phase before the feature, the Day-1 phase after it, and Day 2 without any feature, by arm (bars: mean of the five sessions; dots: sessions, connected). Within each arm, Day 2 does not differ from the Day-1 post-feature phase (paired t over sessions); across arms, every note-arm session is calmer than its paired control session on Day 2.

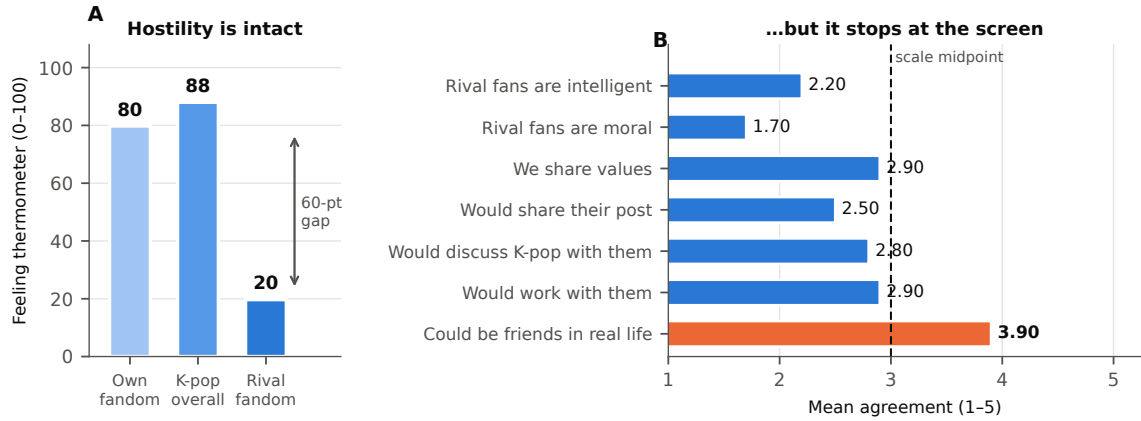


Fig. 6. The ceasefire did not make them like each other, but the dislike is confined to the platform (pilot cohorts, $n = 10$). (A) Feeling thermometer toward one's own fandom, K-pop fans overall, and the rival fandom: the shared identity is warm, the rival is cold. (B) Agreement items about rival-fandom members on a 1-5 scale: they are rated as unintelligent and immoral, yet the single item participants endorsed most was that a rival fan could be a friend in real life.

survey export pending: add before/after change in the thermometer gap, the K-pop thermometer and the contact items, and the two feature items (note changes the atmosphere; fandom still matters).] The enemy in the thread, in other words, is not an enemy in the room—a distinction participants also drew unprompted in the focus groups. [TODO: add the focus-group quote on online versus real-life enemies.]

Table 3. Post-session survey outcomes for treatment-arm participants of the pilot cohorts ($n = 10$). Scale items are 1–5; thermometers are 0–100. Participants experienced the sessions as heated, granted the note little legitimacy, reported high reactance, and their outgroup attitudes stayed cold.

Measure	M (SD)
Legitimacy of the note	2.20 (0.89)
Reactance toward the note	4.25 (0.71)
Perceived similarity to rival fandom	3.02 (1.70)
Perception of rival fandom (intelligent, moral)	1.95 (0.90)
Cross-fandom contact intentions	3.02 (0.79)
Session felt heated / attacked	4.20 (0.79)
Thermometer: own fandom	79.8
Thermometer: rival fandom	19.7
Thermometer gap (own – rival)	60 (43.0)

Survey responses and focus-group accounts pointed in different directions. While participants rated the note low in legitimacy, they described the pop-out itself as a useful feature: *“This function... well I don’t need to think about blinks too much. I just wrote my own opinion on the note and that’s it. But at least it is useful because I...”* (P#, focus group). [TODO: quote is cut off; complete from transcript and replace P# with the participant code.]

5 DISCUSSION

5.1 A Shared Task Without a Shared Identity

The CIIM framework was necessary, but it worked differently from how we expected. It did not produce a common identity. It produced a shared task that both sides were willing to enter for a short time.

We built the feature to satisfy three requirements: real interdependence (neither side can publish alone), dual identities kept visible (badges remain), and no explicit common-identity framing (prompts address K-pop fans without asking anyone to identify as one group).

Behavior changed, but recategorization did not occur. Outgroup similarity stayed at the scale midpoint (3.02/5), the own–rival thermometer gap remained at 60 points, and perception of the rival fandom stayed low (1.95/5). Participants also rated the note low in legitimacy (2.20/5) and high in reactance (4.25/5). If the mechanism were “we now see them as one of us,” we would expect some loosening in outgroup attitudes or identification. We did not observe it.

The shared task was still doing work, and the control arm shows how much. If interruption alone were sufficient, any pop-up would serve: an announcement, a poll, an image. The control arm received exactly such a feature, matched on salience and cost, and escalated through it by 23.0 points. What the note arm had that the control arm did not was a place for attention to go: each fandom had to write half of a note that could not publish until both halves existed, and a third of the arm’s messages after the feature were about doing so. Without a superordinate task both sides are willing to enter, participants have nowhere to go: they either ignore the interruption or read it as one more position and absorb it into the argument.

The common-identity framing and the pause are therefore layered rather than alternative. The framing supplies somewhere to go once the argument is left; the pause is the step that does the work, and it stops short of recategorization. Moving attention was enough.

This also accounts for low legitimacy alongside a prevented escalation. Participants did not accept the note’s authority over their dispute, but they did accept the task of writing something as a K-pop fan. The objects of judgment differ: the shared identity supplied a reason to participate, not an outcome of identification.

One alternative reading should be kept. A drop in “they” could look like common identity, but a change of subject produces a similar linguistic pattern, and the drop here was matched by a rise in naming the rival fandom rather than by any rise in “we”: participants did not stop talking about the other side; they stopped pointing at it. That fits attention moving to a task in which the other side is a named co-author better than it fits recategorization, and we treat it as parallel to the attention account rather than as evidence for CIIM.

5.2 Why a Pause Can Hold

Escalation requires sustained attention. In our seeded sessions toxicity did not start high—both arms opened near 29.1 and climbed by about a point a minute—but it did need attention to keep climbing. Once the note pulled attention elsewhere, the climb stopped; in the control arm, where nothing pulled, it continued for the rest of the hour.

Interrupting attention may therefore be enough to interrupt escalation, without changing what anyone believes. Depolarization interventions typically target beliefs, attitudes, group boundaries, or perceptions of the outgroup. Community Notes competes for none of these. It competes for where attention sits in the moment, which is a lower bar: participants need neither accept another position nor come to like the other side. But not every interruption will do. Attention needs somewhere to land—something both sides are willing to enter and must complete together. This is what the shared task provides. It does not change who participants are; it gives them somewhere to go for a few minutes.

Attention is itself scarce on platforms. Recommendation, reply, and engagement mechanics keep users inside conflicts [29], and participants themselves attributed sustained hostility to platform and industry incentives. An attention-based intervention need not compete with those mechanics over positions. It needs to win a few minutes.

What carried over to Day 2 was a level, not an attitude. We expected an attention effect to be state-dependent and to vanish once the note was gone. Within Day 1 it did decay: by the last window the note arm had drifted back to 58.2. Yet on Day 2, with no feature in either arm, the note arm stayed at its baseline all hour and the control arm started at the level it had reached and stayed there. Neither arm re-ran the climb. The most economical reading is that escalation is path-dependent: a fight that has reached a level resumes at that level, and a fight that was interrupted before it consolidated has nothing to resume. On this reading the note does not buy reconciliation, and it does not buy only minutes either. It buys the difference between a group that has learned how hot its room can get and a group that has not. Whether the level would survive a longer gap, a new topic, or a new session composition is an open question; Day 2 shows only that it survived a night.

5.3 The Intervention Did Not Need to Be Accepted to Work

The escalation was prevented alongside low legitimacy (2.20/5) and very high reactance (4.25/5) in the pilot battery. This is not a contradiction once the objects of resistance and participation are separated. Participants resisted the note’s standing to intervene in their argument. What they accepted was the task of writing something as a K-pop fan, which asked them neither to concede anything to the other side nor to give up their own position. High reactance attached to the note’s authority without fully translating into resistance to the behavioral intervention.

Prior work notes that visible AI invites discounting and reactance [18]. In our design the LLM never appeared as a persuader: participants encountered an interface feature rather than an agent telling them how to talk. This may be one reason resistance did not spread across the whole intervention.

The design implication is that an effective conflict intervention need not be endorsed by its users. Mechanisms that do not depend on goodwill or agreement may be easier to deploy than mechanisms that depend on persuasion, because they need not win consent first. This is consistent with the second point above: competing for attention has a lower bar than competing for belief.

We claim only that high reactance did not prevent the observed behavioral change. We cannot show that reactance is unimportant, nor rule out that higher reactance would defeat the intervention.

5.4 The Note Cooled the Periphery and Heated the Core

Hostility was unevenly produced. One participant per session and arm wrote 42% of all Day-1 messages, and in 9 of the 10 session-arms that participant was BLINK.

These heavy posters are the boundary of the effect, and the boundary is sharper than we expected. In the control arm they escalated with everyone else. In the note arm they went the other way from everyone else: all five became more toxic after the note, by 10.8 to 19.1 points, and wrote a larger share of the arm’s messages than before, while the other 24 participants cooled by -18.2 points on average (Figure 2, right). The arm’s message-level mean is flat only because these two movements cancel. The intervention’s whole effect on toxicity, in other words, came from the periphery, and it came despite the core rather than through it.

The mechanism above explains the periphery. The pause works because a shared task is available for both sides to enter, and those still willing to leave the argument take it. It does not, on its own, explain why the core got hotter. Two readings are open. The heavy posters may have experienced the note as a provocation, an outside party intervening in a fight they were winning, and answered it; the pilot battery’s high reactance is consistent with this. Or the departure of the periphery may have left them a smaller, more concentrated audience of the other side’s heavy poster, so that what remained of the thread was the fight at its purest. The export carries no message text or reply structure, so we cannot yet tell these apart. **[TODO: check the heavy posters’ post-note messages in the raw log: are they replying to the note, or to the rival heavy poster?]**

For cooperative interdependence, this suggests that online cooperative mechanisms reach those still willing to participate more readily than those already committed to the fight, and may cost something at the core. This limits where the mechanism applies rather than undermining it, but it changes what a deployment should watch: an arm-level mean can be flat while the room has split.

5.5 Design Implications

Four implications follow for platforms that host rival communities. *Design ceasefires, not conversions.* An intervention can stop an escalation without changing minds, so its success should be measured in behavior over time rather than in attitude change, and it need not be liked, or even accepted as legitimate, to work. *Give attention somewhere to go.* Interruption alone is not enough—the control arm’s inert feature interrupted nothing—so the pause has to land on a task both sides are willing to enter and neither can finish alone, which is what the two-slot note supplies. *Intervene before the room learns its temperature.* The level a session reached on Day 1 was the level it resumed on Day 2, so the value of a pause lies in when it comes; a trigger that fires while the climb is still in progress, as ours did, prevents a level from being set rather than lowering one already set. *Watch people, not the room.* The note cooled the periphery and heated

the core, and the arm-level mean showed neither, so a trigger and a dashboard that follow individual trajectories—who is escalating, who is being drawn in, who is left in the thread once the others have gone—would both deploy the pause where it can still catch someone and reveal the cost it may impose on those it cannot. Finally, participants’ own line between the enemy on screen and the person off it (Figure 6) suggests a design target we have not yet pursued: features that make the off-screen person visible inside the thread.

5.6 Limitations

Six limitations bound these results. First, the unit of inference is the session, and there are five per arm; the difference-in-differences is consistent across all five, but the paired tests over sessions rest on four degrees of freedom, and the message-level model treats messages as independent within sessions, which they are not. Second, the message log analyzed here has had its text removed and carries no reply structure, so the topic and pronoun codes are the only window on what was said, and the reading of the heavy posters’ behavior in Section 5.4 cannot yet be checked against their words. Third, the survey battery available to us was collected in the pilot cohorts rather than in the five sessions reported; attitude stability is therefore inferred from how pilot participants received the note, not from a pre-post comparison in the sessions whose behavior we report, and the sessions’ own battery must replace it. Fourth, the heaviest poster was BLINK in 9 of 10 session-arms, so the core-versus-periphery contrast is partly confounded with fandom; we cannot say whether the core heated because it was the core or because of who was in it. Fifth, the composition of the platform’s 0–100 toxicity score and the reliability of the three message codes are not yet reported. Sixth, persistence is shown for one night: whether the level a session reaches survives a longer gap or a change of composition is untested. Two further comparisons await data not yet exported: the sessions’ own pre-survey and post-survey battery, and quantitative coding of the focus-group transcripts against the behavioral record.

6 CONCLUSION

We built a platform feature that operationalizes cooperative interdependence: a Community Note that members of rival fandoms must co-author before it can publish, assembled backstage by an LLM that never enters the conversation. In five two-day sessions between ARMY and BLINK members, matched control sessions escalated by 23.0 points after an inert feature appeared and note sessions did not, a 25.3-point difference that held in every session and was still there the next day with no feature on screen. The calm was not evenly bought: the periphery of each fight cooled while its heaviest poster grew hotter, and references to the rival fandom as “they” gave way to references by name. The pattern is consistent with an attention account: the co-authored note gives an escalating fight somewhere else to go, without asking anyone to change what they believe or who they are. What it bought was a ceasefire rather than a peace—and the participants themselves reported that the enemy in the thread would not be one in real life. A pause, it turns out, can be built, and it can hold.

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